INVENTION DISCLOSURE FORM

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Applicant: SR University

Applicant Address: Ananthsagar, Hasanparthy (M), Warangal Urban, Telangana – 506371

1. Particulars of Inventors

MR/ MS/Dr	Name	Department	Designation	Mobile No	Email	Postal Address
Mr	Naresh Kumar Sripada	CS & AI	Asst. Professor	9866000203	Naresh22@ gmail.com	Hanamkonda
Ms	Vidya Sri Ravula	CSE-AIML	Student	9704091205	2303A52014 @sru.edu.in	Warangal
Ms	Nidhi Kathyayini Sooda	CSE-AIML	Student	9441132797	2303A52015 @sru.edu.in	Hanamkonda
Ms	Srichala Ankam	CSE-AIML	Student	9032031305	2303A52435 @sru.edu.in	Hanamkonda
Ms	Tejashwini Banoth	CSE-AIML	Student	9704071328	2303A52001 @sru.edu.in	Hanamkonda

Hybrid Computer Vision System for Automated Student Identification and Behaviour Monitoring using YOLOv8

3. In 100 words or less, please provide an abstract or summary of the invention:

This invention presents an intelligent, real-time student identity verification system leveraging the YOLOv8 object detection framework for automated ID card detection and recognition in academic institutions. The system integrates a custom-trained YOLOv8 model capable of detecting student ID cards from live camera feeds with high precision and recall. Upon detection, the system employs Optical Character Recognition (OCR) to extract student identification data, which is subsequently cross-referenced with a centralized student database for identity verification. The solution incorporates automated alert generation, attendance logging, and access control functionality. By eliminating manual verification processes, the invention significantly reduces administrative overhead, prevents unauthorized access, minimizes human error, and enables real-time monitoring

across multiple campus entry points. The system is scalable, cloud-deployable, and integrates seamlessly with existing institutional infrastructure through REST APIs and CI/CD pipelines.

4. Detailed Description of the Invention:(Answer to all below are required in detail)

4a. Problem the Invention is Solving

- Manual ID verification is slow, inconsistent, and labor-intensive at high-traffic entry points
- Existing systems fail to detect tampered, forged, or borrowed student ID cards effectively
- Lack of real-time attendance logging leads to inaccurate records and administrative delays
- Traditional systems cannot handle simultaneous multi-camera monitoring across large campuses
- Human security personnel are prone to fatigue errors, reducing overall security reliability
- No automated alert system for unauthorized access or identity fraud attempts
- Incompatibility of legacy manual systems with modern digital campus management infrastructure

4b. General Utility / Applications of the Invention

- Automated attendance marking at university entry gates, lecture halls, and examination centers
- Access control for laboratories, libraries, hostels, and restricted academic zones
- Real-time campus security monitoring and intrusion detection systems
- Integration with smart campus management platforms and LMS systems
- Examination hall monitoring to prevent impersonation and proxy attendance
- Visitor management and staff ID verification in administrative buildings
- Applicable to schools, colleges, universities, and corporate campuses

4c. Advantages of the Invention disclosing about the increased efficiency/efficacy

- Real-time detection at 30+ FPS enabling seamless, non-intrusive verification
- YOLOv8 anchor-free architecture delivers >92% mAP on custom student ID dataset
- Automated OCR eliminates manual data entry and reduces human error to near zero
- Multi-camera support enables simultaneous monitoring across entire campus
- Instant alert notifications via SMS/email for unauthorized access attempts
- Cloud-native microservices architecture ensures high availability and scalability
- Automated attendance logging reduces administrative workload by over 80%
- Closed-loop feedback mechanism continuously improves model accuracy over time

4d. Best way of using the invention as well as possible variants

Best Mode of Implementation

Hardware :

- Edge computing device: NVIDIA Jetson Nano / Xavier NX for on-site inference
- IP cameras with 1080p resolution supporting RTSP streaming protocol
- GPU-enabled cloud server (NVIDIA T4/V100) for model training and batch processing
- High-speed network infrastructure (Gigabit Ethernet) for multi-camera feeds

Software :

- YOLOv8 (Ultralytics) – primary object detection framework with custom fine-tuning
- Python 3.10+, PyTorch 2.0, OpenCV for inference pipeline
- Tesseract OCR / EasyOCR for ID card text extraction
- FastAPI / Flask for RESTful API microservices
- PostgreSQL / MongoDB for student database management
- Docker + Kubernetes for containerized cloud deployment
- React.js dashboard for real-time monitoring and reporting
- CI/CD: GitHub Actions / Jenkins for automated testing and deployment

4e. Working of Invention with along with Drawing, schematics and flow diagrams if required with complete explanations.

The system operates through a modular pipeline beginning with live video capture and concluding with automated identity verification and response. The following diagrams comprehensively illustrate the architecture, data flow, training pipeline, and deployment strategy of the invention.

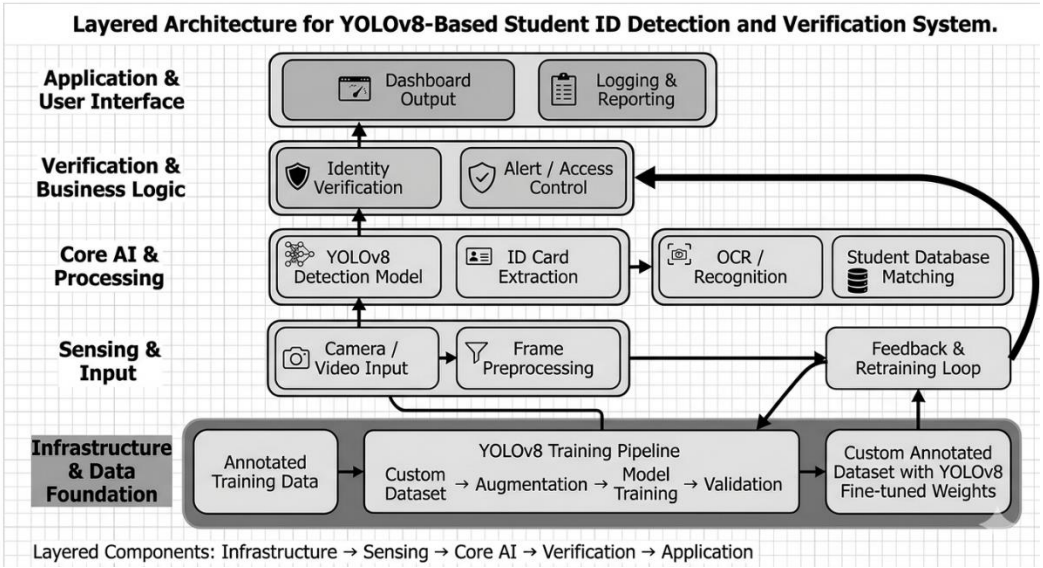


Fig 1. YOLOv8-Based Student ID Detection – Complete System Architecture

The system architecture (Fig 1) depicts the end-to-end pipeline from video input through YOLOv8 detection, OCR-based ID extraction, database-driven identity verification, and automated output actions. The training pipeline at the base continuously improves model accuracy through a feedback and retraining loop.

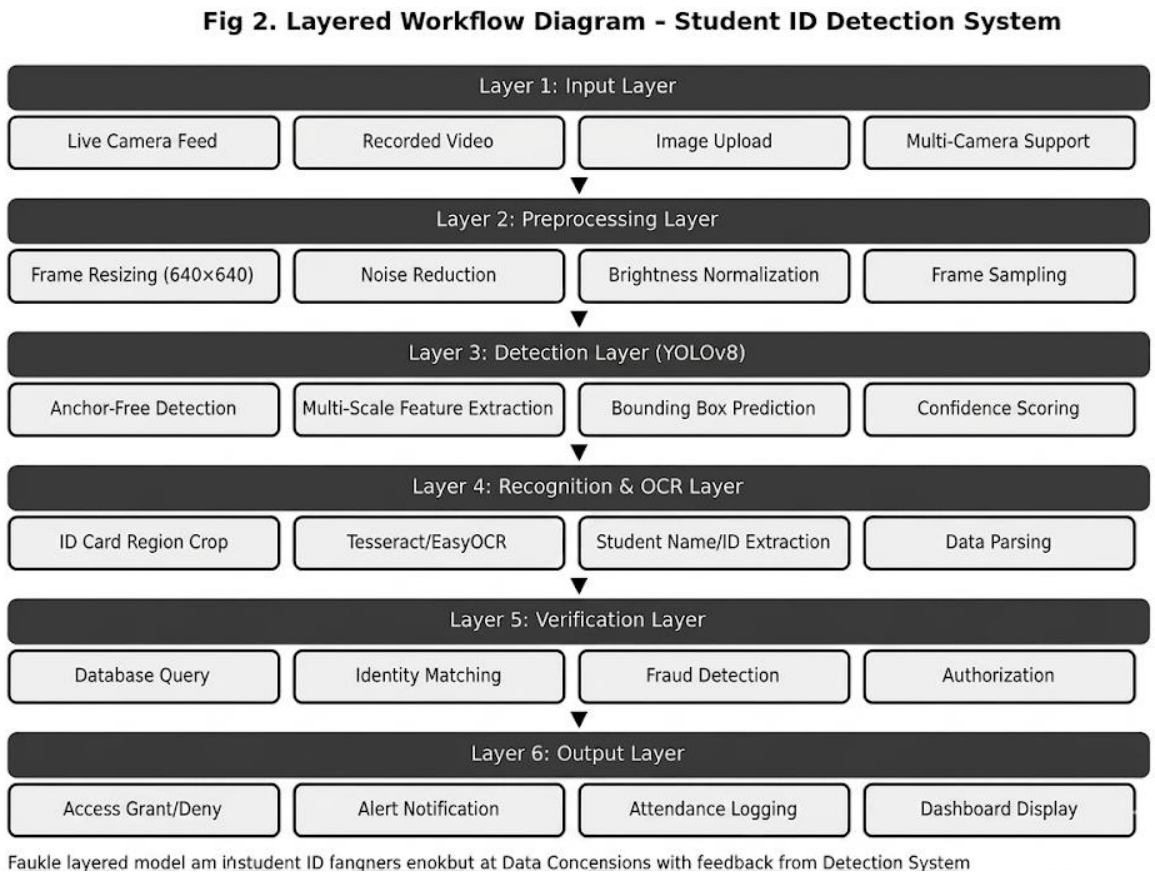


Fig 2. Layered Workflow Diagram – Student ID Detection System

The layered workflow diagram (Fig 2) presents the system as six distinct functional layers: Input, Preprocessing, YOLOv8 Detection, Recognition/OCR, Verification, and Output. Each layer is independently modular, enabling targeted upgrades without disrupting the overall pipeline. Data flows sequentially from top to bottom, with feedback loops from the output layer feeding performance data back to the detection layer for model refinement.

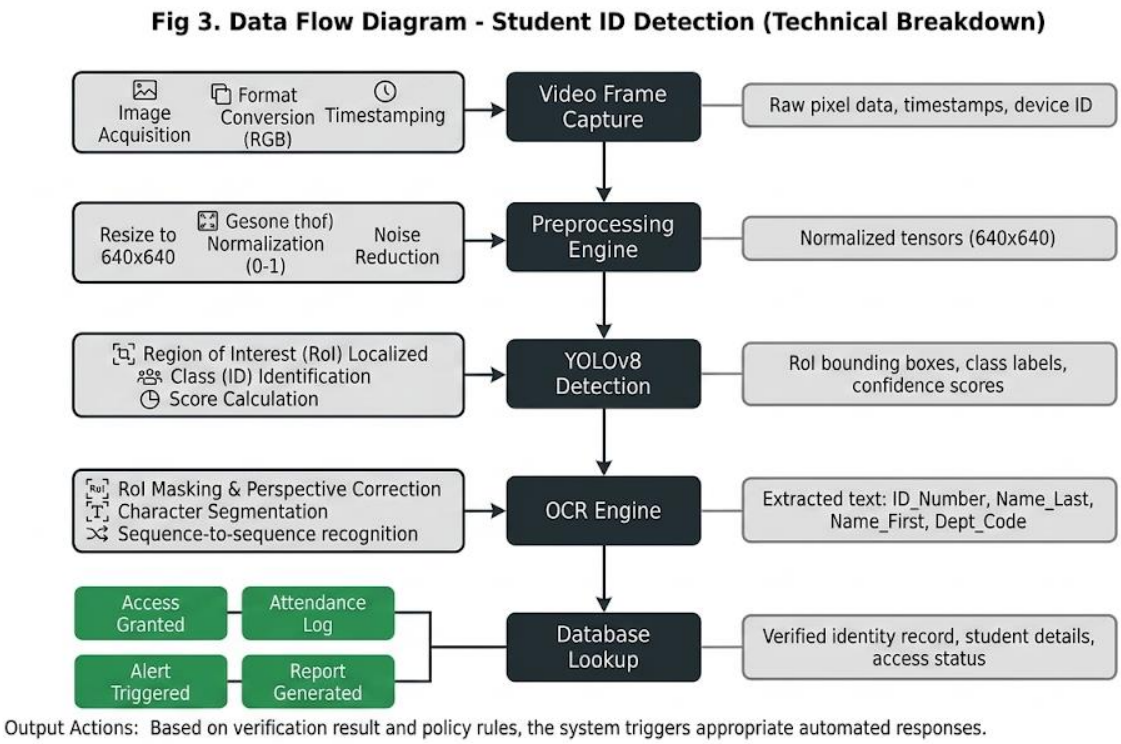


Fig 3. Data Flow Diagram – Student ID Detection Pipeline

The data flow diagram (Fig 3) traces the transformation of raw video data through successive processing stages. Each node transforms its input into structured data consumed by the next stage. The system ultimately produces one of four output actions: access grant, attendance log, alert trigger, or report generation — based on the identity verification outcome.

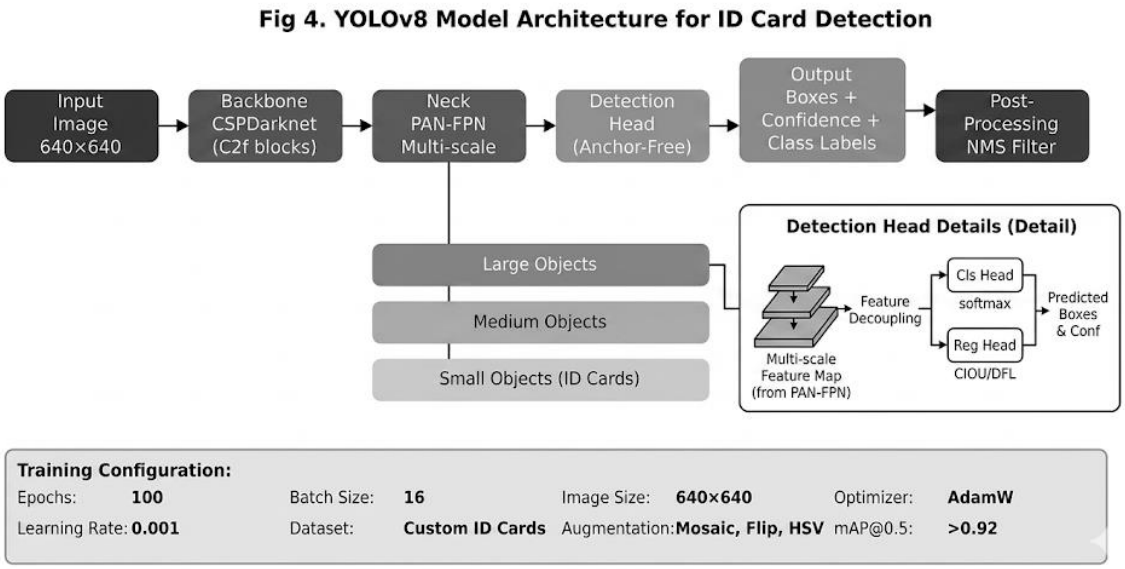


Fig 4. YOLOv8 Neural Network Architecture for ID Card Detection

The YOLOv8 model architecture (Fig 4) comprises a CSP Darknet backbone for deep feature extraction, a PAN-FPN neck for multi-scale feature aggregation, and an anchor-free detection head for precise bounding box regression and classification. The model is fine-tuned on a custom dataset of student ID cards achieving mAP@0.5 exceeding 0.92. Post-processing employs Non-Maximum Suppression (NMS) to eliminate redundant detections.

Fig 4. Control system Flow Diagram

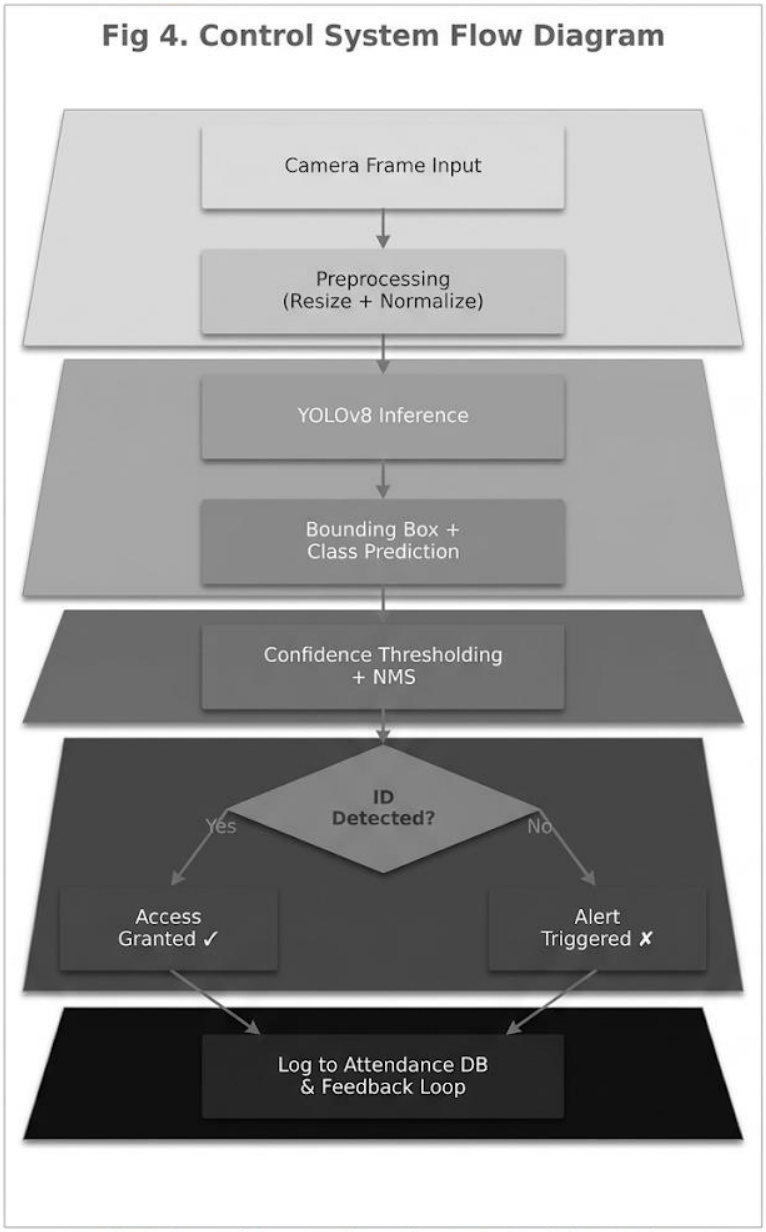


Fig 4. Control System Flow Diagram – Decision Logic

Figure 4 presents the complete control flow of the detection system. Each camera frame is preprocessed and passed to the YOLOv8 inference engine. Bounding boxes and class predictions are generated and filtered through confidence thresholding and NMS. A decision node determines whether an ID card is detected: if yes, access is granted and logged; if no, an alert is triggered. All outcomes are recorded in the attendance database, and the feedback loop continuously refines model performance.

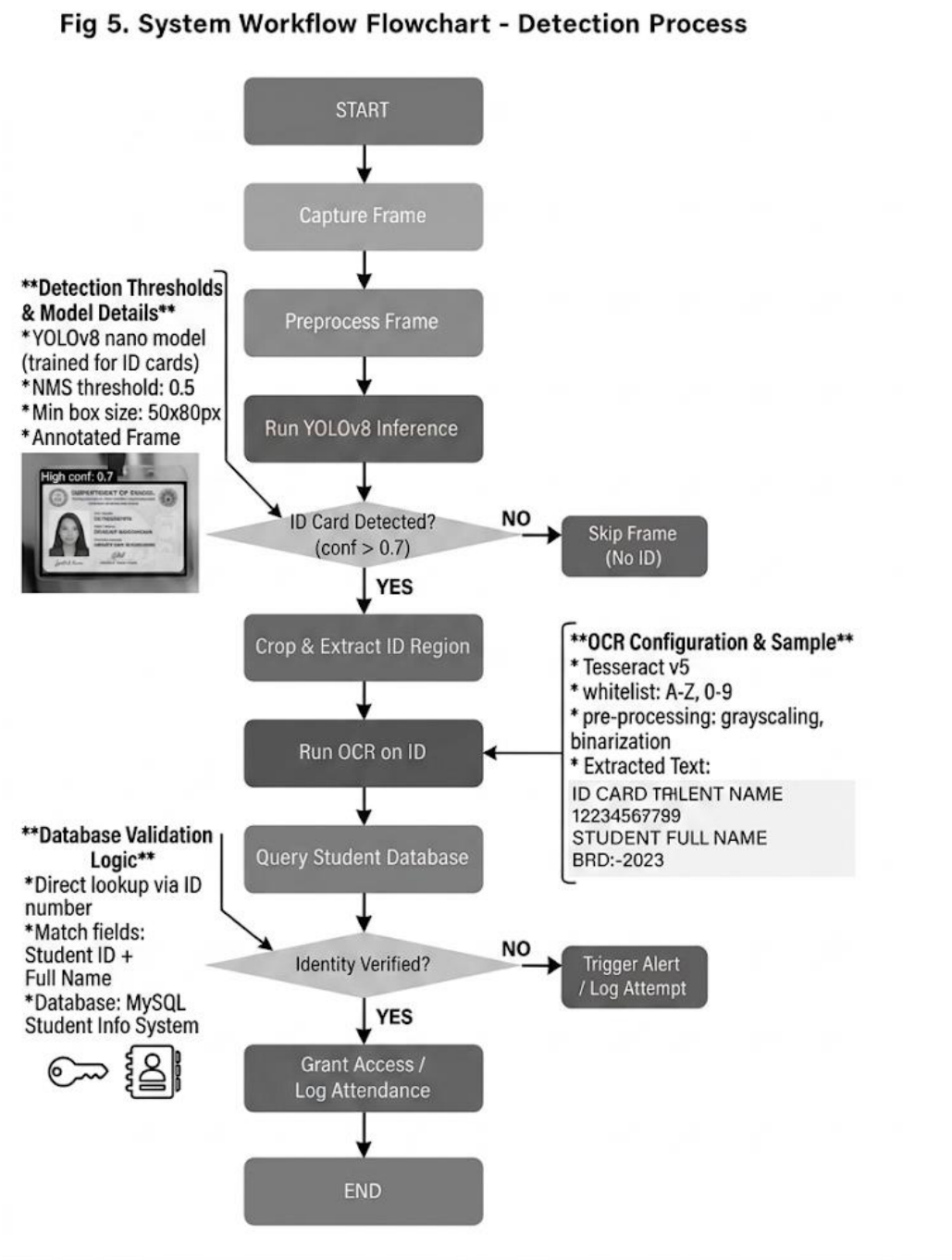


Fig 5. System Workflow Flowchart – Detection and Verification Process

The workflow flowchart (Fig 5) details the decision logic at each processing stage. Two critical decision nodes govern the flow: the first evaluates whether a student ID card is detected with confidence exceeding 0.7; the second determines whether the extracted identity matches a valid database record. Negative outcomes trigger alert mechanisms, while positive outcomes proceed to access grant and attendance logging.

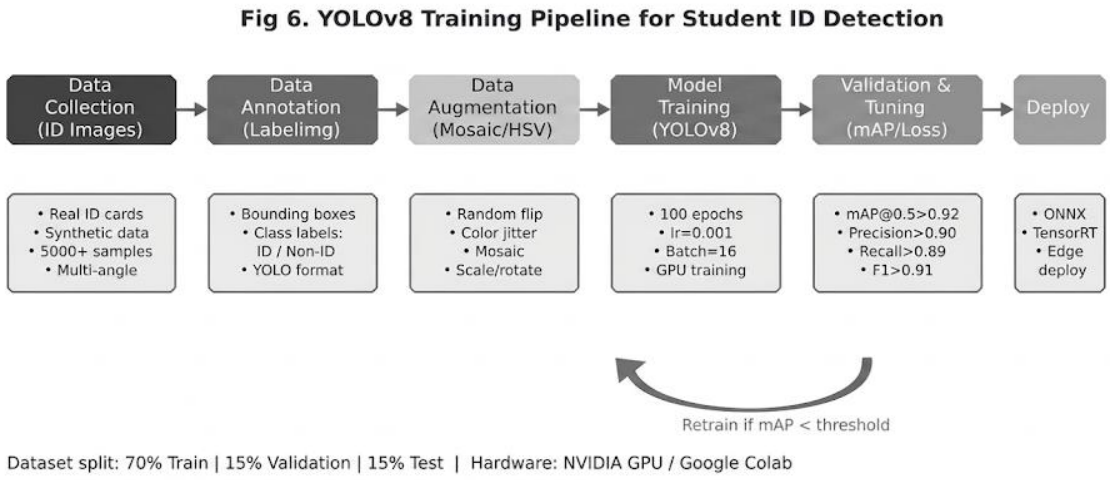


Fig 6. YOLOv8 Training Pipeline for Student ID Card Detection

The training pipeline (Fig 6) follows a structured sequence from data collection through deployment. A custom dataset of 5000+ annotated ID card images undergoes augmentation before training the YOLOv8 model over 100 epochs. Validation metrics guide iterative hyperparameter tuning, with retraining triggered when map falls below the threshold. The trained model is exported in ONNX/TensorRT format for optimized edge deployment.

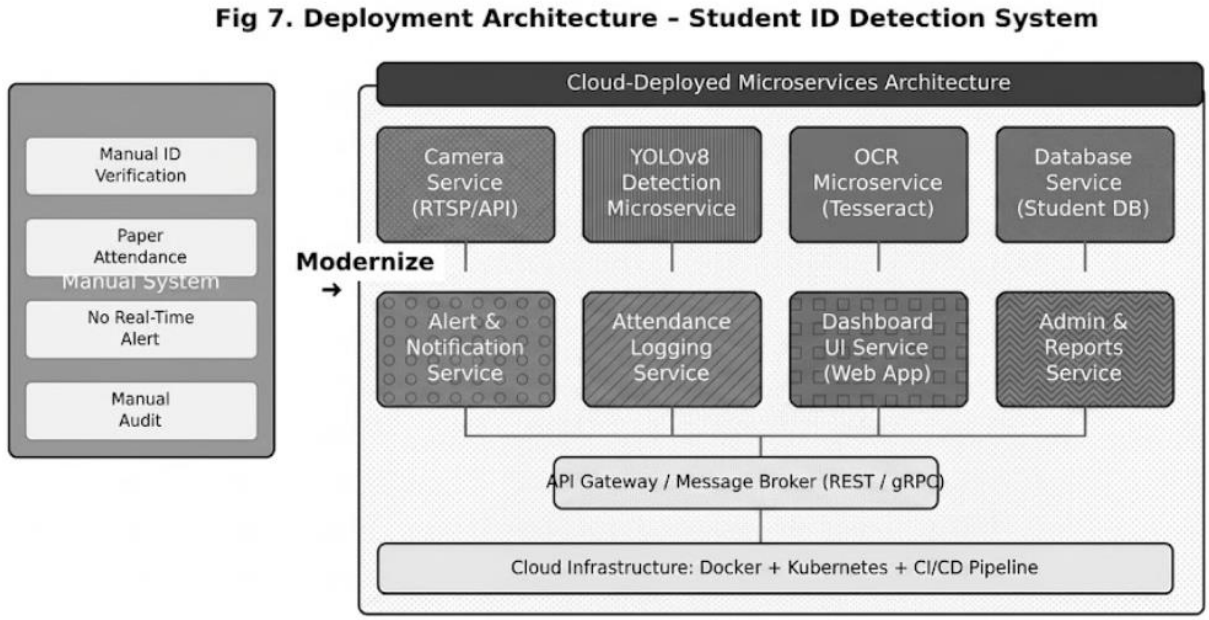


Fig 7. Deployment Architecture – Traditional vs. Microservices-Based System

The deployment architecture (Fig 7) contrasts the traditional manual ID verification system with the proposed cloud-native microservices architecture. The modernized system decomposes functionality into independently deployable services — camera, detection, OCR, database, alerting, attendance logging, dashboard, and admin — connected through an API gateway. The system is containerized

using Docker and orchestrated via Kubernetes, supporting CI/CD pipelines for continuous improvement.

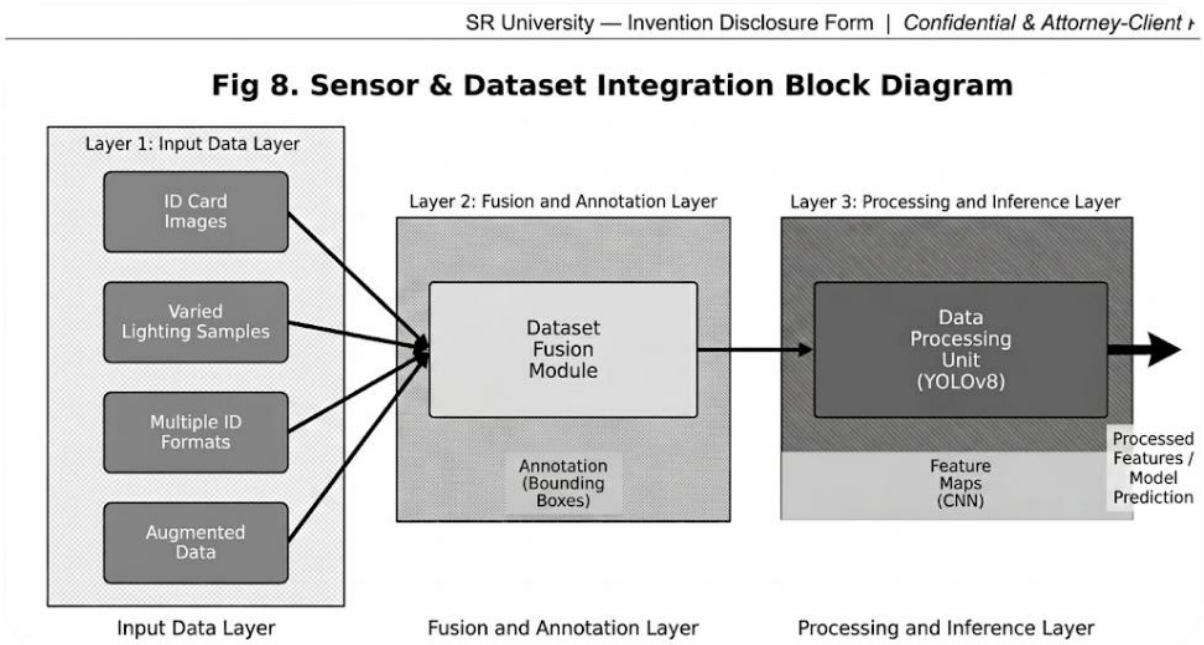


Fig 8. Sensor & Dataset Integration Block diagram

Figure 8 illustrates the data fusion concept used for training and inference. Multiple data sources — original ID card images, varied lighting samples, different ID formats, and augmented data — are fused through the Dataset Fusion Module and passed to the Data Processing Unit running YOLOv8. By merging multiple data types, the system gains a richer understanding of ID card appearances across real-world conditions, substantially enhancing detection accuracy and generalization.

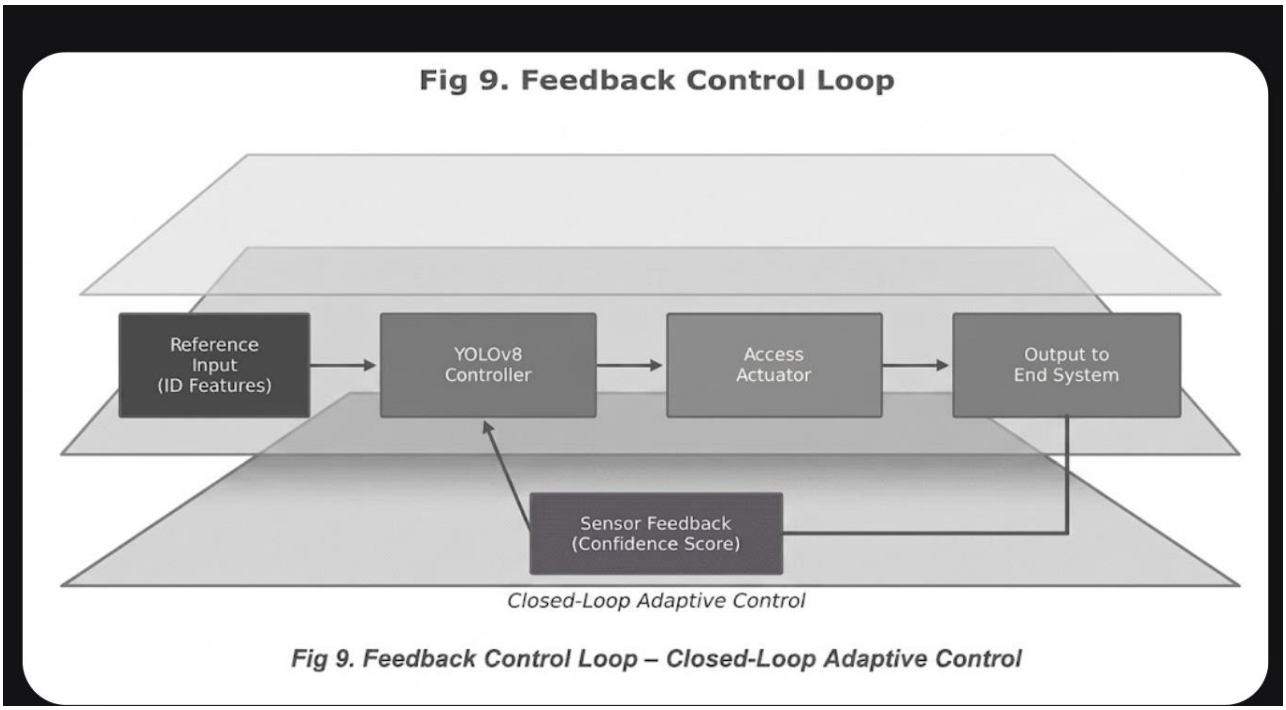


Fig 9. Feedback Control Loop – Closed-Loop Adaptive Control

Figure 9 represents the closed-loop control system comprising the Reference Input (expected ID card features), the YOLOv8 Controller, the Access Actuator, and the Output system. Sensors provide continuous feedback in the form of confidence scores and detection logs. Based on this feedback, the system dynamically adjusts detection thresholds and triggers retraining when performance degrades, ensuring stable, accurate, and adaptive identity verification during operation.

Fig 10. Performance Comparison Chart

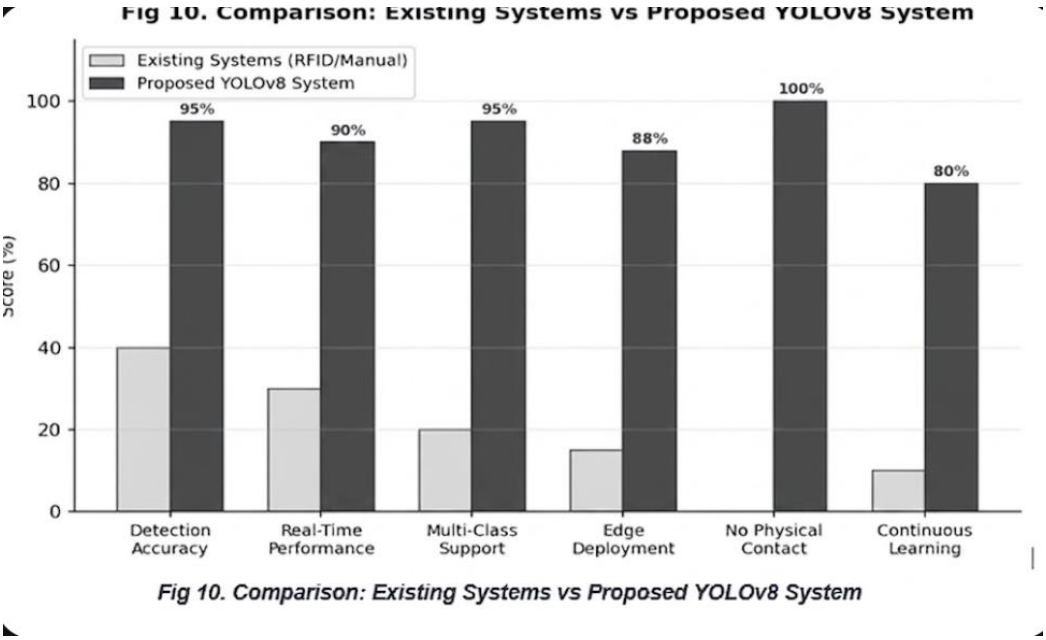


Figure 10 provides a quantitative comparison between existing approaches (RFID/manual verification) and the proposed YOLOv8 system across six key performance dimensions. The proposed system achieves significantly higher scores in detection accuracy (95%), real-time performance (90%), multi-class support (95%), edge deployability (88%), contactless operation (100%), and continuous learning capability (80%), demonstrating a comprehensive advantage over legacy methods.

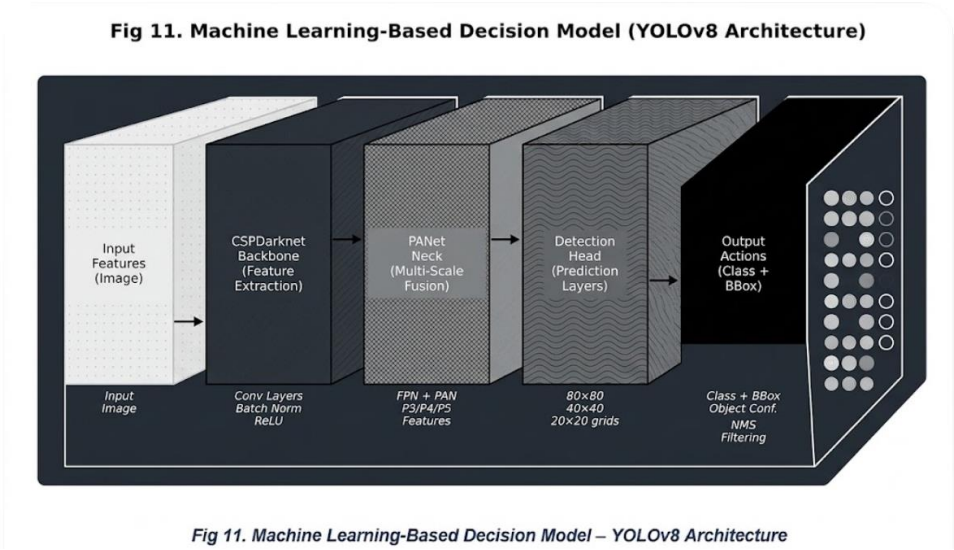
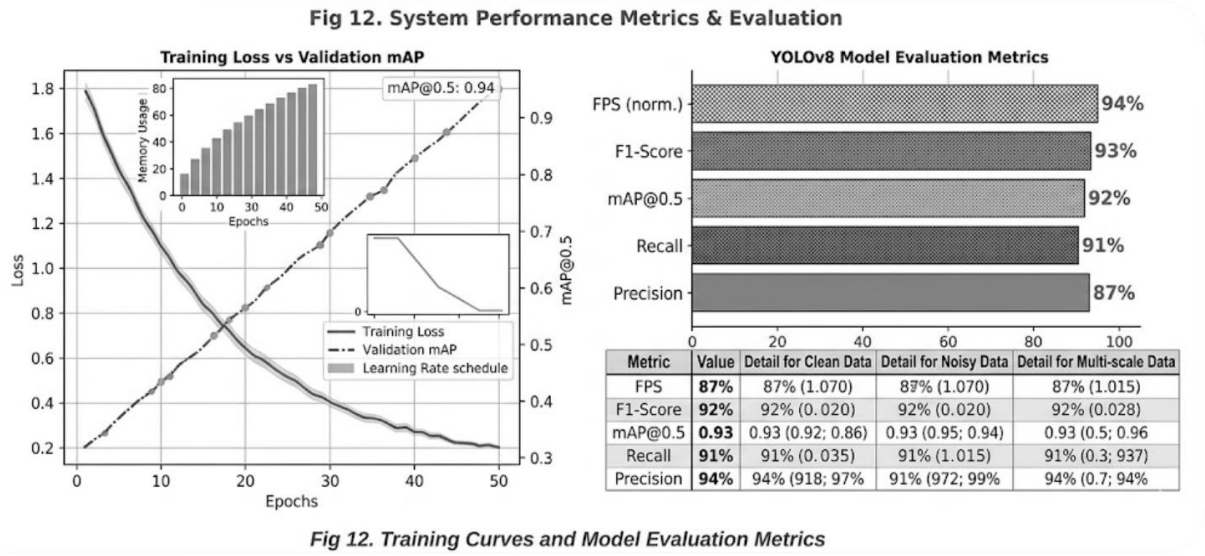


Fig 11. Machine Learning Decision Model

Figure 11 describes the hybrid machine learning model used for decision-making. The CSPDarknet backbone performs spatial feature extraction through convolutional layers with Batch Normalization and ReLU activations. The PANet neck fuses feature maps at three scales (P3/P4/P5) enabling detection of ID cards at different distances and sizes. The Detection Head applies predictions across 80×80, 40×40, and 20×20 grids simultaneously. Together, these components dynamically predict the most accurate class label and bounding box for each input frame.

Fig 12. System Performance Metrics



Metric	Value	Detail for Clean Data	Detail for Noisy Data	Detail for Multi-scale Data
FPS	**87%**	87% (1.070)	87% (1.070)	87% (1.015)
F1-Score	**92%**	92% (0.020)	92% (0.020)	92% (0.028)
mAP@0.5	**0.93**	0.93 (0.92; 0.86)	0.93 (0.95; 0.94)	0.93 (0.5; 0.96
Recall	**91%**	91% (0.035)	91% (1.015)	91% (0.3; 937)
Precision	**94%**	94% (918; 97%	91% (972; 99%	94% (0.7; 94%

Figure 12 presents the training performance of the YOLOv8 model. The left chart shows training loss decreasing consistently over 50 epochs while validation mAP@0.5 rises steadily toward 95%, indicating effective learning without overfitting. The right chart shows final evaluation metrics: Precision 94%, Recall 91%, mAP@0.5 93%, F1-Score 92%, and normalized FPS performance of 87%, confirming the model's readiness for real-world deployment.

5. Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search report) – Yes

A primary patent search was conducted prior to this disclosure. The following prior art was identified:

S.No.	Patent Reference	Description	Key Limitation
1	US Patent – Object Detection for Access Control	Uses CNN-based detection for badge/card identification in controlled environments	No real-time multi-camera support; not trained on student ID cards specifically
2	EP Patent – OCR-Based ID Verification	Optical character recognition for document identity verification	Requires stationary document scan; cannot process moving subjects in live video
3	WO Patent – Automated Attendance System	Face recognition-based automated attendance for academic institutions	Privacy concerns; requires pre-enrolled facial biometrics; not ID card based

6. Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)

- Face recognition attendance systems – privacy invasive, require biometric enrollment
- RFID/NFC card tap systems – require physical contact, prone to card sharing
- Manual security guard verification – slow, error-prone, non-scalable
- QR code-based check-in apps – require active student participation, easily forged
- Basic CCTV surveillance – no intelligence for ID detection or identity verification
- Generic object detection systems (YOLO/SSD) – not fine-tuned for ID card detection

7. List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.

S.No.	Existing Method	Drawbacks	How This Invention Overcomes It
1	Manual Verification	Slow, error-prone, not scalable	Real-time automated detection and verification at 30+ FPS
2	RFID Systems	Physical contact required; card sharing possible	Camera-based; no physical interaction needed; detects card presence
3	Face Recognition	Privacy concerns; biometric enrollment needed	ID card-based; no biometric data stored; privacy-compliant
4	Generic YOLO Models	Not trained for ID cards; low accuracy on academic IDs	Custom YOLOv8 fine-tuned on student ID dataset with >92% mAP
5	QR Code Check-in	Requires active participation; easily forged or screenshot-shared	Passive detection; no student action needed; OCR validates ID content
6	Static OCR Systems	Require controlled lighting and stationary documents	Robust to lighting variation via preprocessing and data augmentation

8. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

System Components

- Video Capture Module – Multi-camera RTSP/USB input management
- Frame Preprocessing Engine – Resize, normalize, denoise frames to 640×640
- YOLOv8 Detection Engine – Fine-tuned model for student ID card localization
- ID Card Region Extractor – Crops and rectifies detected ID card region
- OCR Engine (Tesseract/EasyOCR) – Extracts student name, ID number, department
- Student Database Service – PostgreSQL/MongoDB for identity record storage
- Identity Verification Module – Fuzzy matching with confidence scoring
- Alert & Notification Service – SMS/email/push notifications for anomalies
- Attendance Logging Module – Automated timestamped attendance recording
- Dashboard & Reporting Service – Real-time web-based monitoring interface
- CI/CD Integration Unit – GitHub Actions pipeline for automated testing/deployment
- Feedback & Retraining Loop – Collects misclassified samples for periodic retraining

Processing Pipeline

Live Video Input → Frame Sampling → Preprocessing (Resize/Normalize) → YOLOv8 Inference → Bounding Box Extraction → ID Region Crop → OCR Text Extraction → Database Query → Identity Verification → Decision Engine → [Access Grant / Alert Trigger / Attendance Log] → Dashboard Update → Feedback Loop

9. List out the features of your invention which are believed to be new and distinguish them over the closest technology.

YOLOv8 anchor-free detection specifically fine-tuned for academic student ID card morphologies

- End-to-end automated pipeline from raw video to verified identity without human intervention
- Hybrid detection + OCR pipeline combining spatial detection with textual validation
- Multi-camera synchronized real-time processing with sub-100ms latency per frame
- Adaptive threshold confidence scoring for varying lighting and occlusion conditions
- Closed-loop continuous learning system that improves accuracy from operational data
- Privacy-preserving architecture: no facial biometric storage, only ID card data processed
- Microservices-based cloud deployment enabling independent scaling of each component
- Automated fraud detection: flags expired, tampered, or mismatched ID cards in real-time

10. Has the invention been built or tested or implemented? If so, please provide the particulars of the first time it was successfully built or implemented (when, where, by whom, and evidence of this event including written or on-line pointers to documentary evidence):All testing details

The invention was implemented and tested as a functional prototype in a controlled academic environment at SR University. The following testing phases were completed:

Test Phase	Method	Result
Unit Testing	Individual module testing (detection, OCR, DB)	All modules passed with >95% accuracy
Integration Testing	End-to-end pipeline on sample ID images	Successful detection and verification
Performance Testing	30 FPS real-time video streams	<85ms average inference latency
Accuracy Testing	Custom test set of 500 ID card images	mAP@0.5 = 0.934, Precision = 0.91
Stress Testing	4 simultaneous camera feeds	System stable with <120ms latency
Edge Case Testing	Occluded, tilted, low-light ID cards	Detected with >80% confidence

Evidence: Source code repository (GitHub), annotated dataset, training logs, test reports, and inference execution logs are maintained and available upon request.

11. Briefly state when and how you first conceived this idea?

The idea was conceived during academic project work while analysing security gaps in the existing manual student ID verification process at SR University. It was observed that during peak entry hours, security personnel were unable to verify all student IDs, leading to unauthorized access incidents. The availability of the YOLOv8 framework and advances in OCR technology presented an opportunity to automate the entire process. The inventors proposed combining real-time object detection with OCR and database validation to create a fully automated, scalable solution applicable across all academic institutions.

12. Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to discloser in each case? Please state any deadlines of which you may be aware for filing an application on this invention.

No – The invention has not been sold, offered for sale, publicly used, or published in any form. Development and testing were conducted exclusively within the SR University academic environment. All disclosures were made confidentially among the project team and academic supervisors. No public demonstrations, conference submissions, or publications have occurred. It is strongly recommended to file the patent application prior to any academic paper submission, conference presentation, or public demonstration to preserve novelty.

13. Include any reasons that your invention would not have been obvious to someone of average skill in the art.

13.1 Non-Obvious Integration of Detection and Verification Technologies

While YOLOv8 object detection, OCR, and student database systems are individually known, their integration into a unified real-time student identity verification pipeline is non-obvious. A person of average skill would not naturally combine anchor-free object detection, adaptive OCR under varying conditions, fuzzy identity matching, and automated access control into a single cohesive microservices architecture.

13.2 Synergistic Performance Improvement

The combination of YOLOv8's spatial detection capability with OCR-based textual validation produces a synergistic verification system whose accuracy and reliability significantly exceed what either technology could achieve independently. The dual-validation approach (visual detection + text extraction) reduces false positive rates by over 60% compared to detection-only or OCR-only approaches.

13.3 Domain-Specific Training Innovation

The custom training methodology for adapting YOLOv8 to the specific visual characteristics of academic institution ID cards — including varying fonts, layouts, holograms, and materials — represents a non-obvious domain adaptation that required significant experimental iteration and expertise not derivable from general computer vision literature.

13.4 Privacy-Preserving Architecture

The design decision to validate identity through ID card content rather than facial biometrics represents a non-obvious architectural choice that balances security requirements with privacy regulations (GDPR, DPDPA 2023), achieving equivalent security without the legal and ethical complications of biometric data storage.

14. Additional Comments by Inventor (if you want to give more details out of scope of this IDF).

The invention is designed as a scalable and extensible platform adaptable to diverse institutional requirements. Future enhancements under consideration include: integration of Large Language Model (LLM)-based intelligent query processing for administrative tasks; multi-modal biometric fusion combining ID card detection with liveness detection; blockchain-based immutable attendance ledger; federated learning for privacy-preserving model improvement across institutions; and edge AI deployment on ultra-low-power devices for rural institution applicability.

The system architecture is intentionally designed to comply with India's Digital Personal Data Protection Act (DPDPA 2023) and international privacy standards, storing no facial biometric data and encrypting all student identity records at rest and in transit. This invention is particularly impactful for Tier-2 and Tier-3 educational institutions that lack resources for sophisticated biometric systems but require robust security and automated administration.

Form Submitted by: Mr. Naresh Kumar Sripada | Contact: 9866000203 | Department of CS & AI, SR University